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SYSTEMATIC REVIEW

Overtraining in Resistance Exercise: An Exploratory Systematic Review and Methodological Appraisal of the Literature

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Abstract

Background The balance between training stress and recovery is important for inducing adaptations to improve athletic performance. However, continuously high training loads with insufficient recovery may cause fatigue to accumulate and result in overtraining. A comprehensive systematic review is required to collate overtraining literature and improve the current understanding of the mechanisms underlying functional overreaching (FOR), non-functional overreaching (NFOR) and the overtraining syndrome (OTS) in resistance training.

Objective The objective of this systematic review was to establish markers of overtraining and elucidate the mechanisms underlying maladaptive resistance training conditions. Furthermore, this review aims to critically evaluate the methodological approaches of the overtraining literature.

Methods A systematic literature search was performed on PubMed, Web of Science and SPORTDiscus to identify studies up to June 2019. Electronic databases were searched using terms related to resistance training and overtraining. Records were included if they attempted to induce a state of overreaching or overtraining through resistance exercise in healthy participants.

Results A total of 22 studies were selected for review. Among these studies, eight resulted in decrements in performance and measured changes in performance during a follow-up period. There were four studies that reported decrease in performance yet failed to implement follow-up measures. A total of 10 studies reported no decline in performance. Overall, a lack of standardisation in methodology (follow-up performance testing) and diagnostic criteria prevents consistent determination of FOR, NFOR and OTS in resistance training.

Conclusions Few studies have appropriately established FOR, NFOR or OTS in resistance training. Overtraining may be related to frequent high-intensity and monotonous resistance training. However, no marker other than a sustained decrease in performance has been established as a reliable indicator of overtraining in resistance exercise.

Registration This systematic review was registered on the Open Science Framework (https://osf.io/) (https://doi.org/10.17605/osf.io/5bmsp).

1 Introduction

Many athletes perform resistance exercise or strength training as a means to improve athletic performance. Resistance training programs can be systematically manipulated through changes in training variables (i.e. intensity, volume, exercise selection and sequence, rest intervals, repetition velocity and training frequency) to elicit a specific response (i.e. muscular strength, hypertrophy, endurance, motor performance, balance or coordination) [1, 2]. Strenuous resistance training loads should be applied to obtain high-level performance in strength and power-based sports [2]. The physical performance of an athlete undertaking resistance training may be influenced by their ability to adapt to increasing training loads. Consequently, athletes and coaches are continually challenged to find the optimal balance between training and recovery to enhance athletic performance. A disruption in this balance may result in impaired performance [3, 4].

During a typical training cycle, athletes often undergo periods of overload and recovery in an attempt to optimise physical performance. The process of overload is achieved through an increase in training load [3]. Acute fatigue occurs following overload training which may result in a

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Key Points

Failure to demonstrate decreases in performance and inadequate performance testing during a follow-up period prohibit the diagnosis of FOR, NFOR or OTS in the majority of studies.

Early studies show that frequent and monotonous resistance training performed at high intensity may increase susceptibility to training maladaptation.

The mechanisms that underlie overtraining in resistance exercise remain unclear. Coaches and athletes must continue to rely on training-specific performance decrements to determine FOR, NFOR or OTS.

temporary performance decrement [5]. When adequate rest is prescribed, the resultant acute fatigue may be followed by a positive adaptation or improvement in performance [3]. This process is referred to as functional overreaching (FOR). It is well established that periods of physiological stress are required to stimulate the adaptations responsible for enhancing performance [3]. Accordingly, coaches and athletes often induce FOR as part of a training program to elicit a supercompensation effect. However, continuously high training loads with insufficient recovery may cause fatigue to accumulate and result in long-term reductions in performance capacity. This maladaptation to training can manifest as a state of non-functional overreaching (NFOR) and, in extreme and prolonged cases, may develop into the overtraining syndrome (OTS).

Many previous studies have investigated the mechanisms of overtraining in an attempt to establish indicators of overtraining through the use of various performance, biochemical, physiological and psychological parameters (for review, see Halson et al. [6] and Meeusen et al. [7]). However, despite many attempts, no clear guidelines or diagnostic tools other than a sustained decrease in performance have emerged for the monitoring and detection of FOR, NFOR or OTS in resistance exercise [7]. This may be due to the varied terminology used by previous studies or the difficulty that lies in differentiating between these conditions. This review follows the well-accepted definitions proposed by Meeusen et al. [7], where FOR, NFOR and OTS are differentiated by the length of time required for recovery. FOR is defined as a short-term decrement in performance without severe psychological or other lasting negative symptoms that requires 'days-weeks' to recover. FOR is followed by a subsequent supercompensation effect where performance is improved. NFOR and OTS are caused by overtraining and defined as longer-term decrements in performance with or without a range of other symptoms (e.g. hormonal dysregulation, psychological disturbances, reduced immune function, sleep disorders) that require 'weeks-months' and 'months-…' to recover, respectively [7]. Consequently, the correct diagnosis requires maximal performance testing and can only be made retrospectively. Importantly, the OTS is characterised by an unexplained decline in performance and depends on a wide variety of training and contextual factors. Therefore, the correct diagnosis of OTS should also be based on the exclusion of confounding factors/disorders (e.g. viral disorder, bacterial infection and other physical diseases) [7]. The amount that performance must deteriorate before FOR, NFOR or OTS is diagnosed has been debated in the literature. The magnitude of performance decline required to diagnose FOR, NFOR or OTS may vary widely depending on the specific performance assessment and the training status of the athlete. Despite the fact that a meaningful decrease in sport-specific performance is individual, for the purpose of this review, a statistically significant decline in performance was necessary to indicate a state of possible FOR, NFOR or OTS.

The majority of previous overreaching/overtraining investigations have been conducted in endurance activities [7]. However, previous authors have proposed that overtraining in resistance exercise can elicit considerably different biological responses when compared to overtraining with endurance activities [8, 9]. The mechanisms behind these contrasting responses have not been established. Accordingly, there is a need to investigate overtraining respective to the training mode. It has also been suggested that the aetiology of resistance exercise overtraining differs with regard to whether training volume or intensity is manipulated/increased [7]. However, the mechanisms and manifestation of FOR, NFOR and OTS through resistance exercise remain relatively unknown.

Overtraining has been suggested to be a significant problem for strength or power athletes who undertake primarily resistance training [8]. Determining the point at which training becomes maladaptive continues to challenge both athletes and coaches in their attempts to achieve optimal performance. At present, there is no systematic review of the studies that have examined NFOR, FOR or OTS in resistance-trained athletes. Therefore, the purpose of this systematic review was to critique and appraise the results and methodology of studies that have attempted to induce FOR, NFOR or OTS through resistance exercise in healthy adults. Additionally, the present review aims to elucidate the physiological alterations and potential mechanisms of overtraining, as well as determine if any markers (performance, biochemical, physiological or psychological) may be used to monitor FOR, NFOR or OTS in resistance training.

2 Methods

This review was conducted according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [10]. A systematic review protocol that included the review question, search strategy, exclusion criteria and risk of bias assessment was registered with the Open Science Framework (10.17,605/OSF.IO/5BMSP; 11 January 2019).

2.1 Eligibility Criteria

Eligibility criteria were drafted and subsequently refined by three authors (CG, LW, FI) using a random sample of studies. Included studies met the following study criteria [population, intervention, comparator, outcome, study design (PICOS)] and report characteristics. P: Participants with chronic diseases, metabolic disorders or musculoskeletal injuries were excluded. Studies were considered eligible if they included an intervention aimed to induce FOR, NFOR or OTS through resistance training. I: Interventions were included regardless of whether or not they were able to induce a decrement in performance which is essential for the diagnosis of FOR, NFOR or OTS. Studies involving an acute bout of intensified training (i.e. increased training load) were not eligible for inclusion in this review unless the authors stated that they were attempting to overreach or overtrain participants. Interventions combining resistance training and aerobic training were excluded to account for a possible interference effect. Supplementation studies were eligible for inclusion if the data provided for non-supplemented participants were reported separately. C: All comparators and control conditions were eligible for this review. O: Changes in physical performance following the overtraining period and follow-up were the primary outcome measure of this review. Secondary outcome measures included biochemical, neuro-muscular, psychological and morphologic effects. S: This review considered randomised controlled trials (RCTs), non-randomised controlled trials (NRCTs), uncontrolled trials (UCTs) and grey literature to ensure literature saturation.

2.2 Search

A literature search was conducted by one author (CG) in the following electronic databases: PubMed, Web of Science Core Collection and SPORTDiscus. Databases were searched from inception up until June 2019. No language or publication status restrictions were imposed on the search to ensure literature saturation. The following Boolean search string was used on all databases: (resistance training OR resistance exercise OR strength training OR strength exercise OR weight training OR weight exercise OR weight lifting OR weightlifting) AND (overtrain* OR over-train* OR overreach* OR over-reach* OR fatigue OR underperformance OR underperformance OR under performance OR underrecovery OR under-recovery OR under recovery). A comprehensive search strategy is provided in Table S1. Trial/study registries were searched during a pilot phase; however, due to the non-clinical nature of this review no results were found. In conjunction with the database searches, the reference lists of relevant studies, reviews and books were screened for possible omissions.

2.3 Study Selection

Articles retrieved through the systematic search were exported into a reference management software (EndNote version X8) and all duplicate articles were removed. All references were then imported into Covidence (Covidence Systematic Review Software, Veritas Health Innovation, 2013) for assessment of eligibility. Two authors (CG, NA) independently screened the records by title and abstract, with all potentially eligible references proceeding to full-text screening; conflicts were resolved by a third author (FI). Two authors (CG, NA) then independently screened the full text of all included articles against the eligibility criteria.

2.4 Data Extraction

Data were extracted by two authors (CG, LW) and imported into an Excel spreadsheet that was designed for this review (Table S2). Information extracted from each eligible study included publication details (author, year), participant characteristics (sex, training history), study methods (design, randomisation), overtraining intervention (duration, mode, frequency, volume, intensity, length of follow-up) and comparator. Additionally, pre- and post-intervention values were extracted for primary (physical performance) and secondary (biochemical, physiological and psychological) outcome measures.

2.5 Risk of Bias Assessment

The results between the groups were not of interest for this review; rather the pre-to-post changes were required for assessment of the effects of overtraining. Accordingly, an assessment of bias was undertaken of relevant methodological variables. To comply with the AMSTAR 2 (Assessing the Methodological Quality of Systematic Reviews), assessment of allocation concealment, blinding of participants and assessors was conducted for RCTs. Additionally, confounding and selection bias were assessed for NRCTs and UCTs. Furthermore, risk of bias assessment included a narrative review of the length of follow-up and frequency of performance testing that is required to accurately diagnose a state

of FOR, NOR or OTS. The proposed methods of assessment of risk of bias were changed from the review protocol registered on the Open Science Framework. One author (CG) piloted the proposed method of using the Cochrane risk of bias appraisal tool for the included RCTs. However, due to the nature of this review, the results did not accurately reflect risk of bias in terms of overtraining methodology. For example, if the performance of participants was not assessed during a follow-up period, there would be an inherent risk of bias within the results as an accurate diagnosis of FOR, NFOR or OTS cannot be made.

3 Results

3.1 Study Selection

The initial database search yielded 4649 studies. An additional nine studies were identified through other sources (searching reference lists and consultation with experts) (Fig. 1). Once duplicates were removed, 2992 titles and abstracts were screened for inclusion and of those 2923 studies were excluded based on the eligibility criteria (Sect. 2.1). A total of 69 studies were retrieved as full text and assessed for eligibility and of those 47 were excluded (Table S3). The reasons for exclusion at the full text level are displayed in Fig. 1. Upon completion of these procedures, 22 studies were included for analysis in this systematic review.

3.2 Study Characteristics

Among the 22 included studies, 14 were RCTs, 2 were NRCTs, and 6 were UCTs. Physical performance of participants was measured by 20 of the included studies. Fry et al. [11, 12] reported on secondary outcomes of a previous study by the same author which did include physical performance measures [13]. However, these studies are treated separately throughout this review. At least, one measure of performance significantly declined following the overload program of 12 studies (Fig. 2) [9, 11–21]. Within the studies that reported a decline in a measure of physical performance, eight included follow-up measures after the overload training intervention [11–13, 16–20]. No decline in physical performance was reported in ten of the included studies [22–31]. Lowery et al. [27] did not present performance data for placebo participants and therefore was included in studies that did not induce a decline in performance. Additionally, authors should exercise caution when interpreting the results of Lowery et al. [27] and Wilson et al. [31] due to the controversy surrounding these manuscripts [32]. The descriptive results of the 12 studies that reported decreases in performance are displayed in Table 1. The descriptive

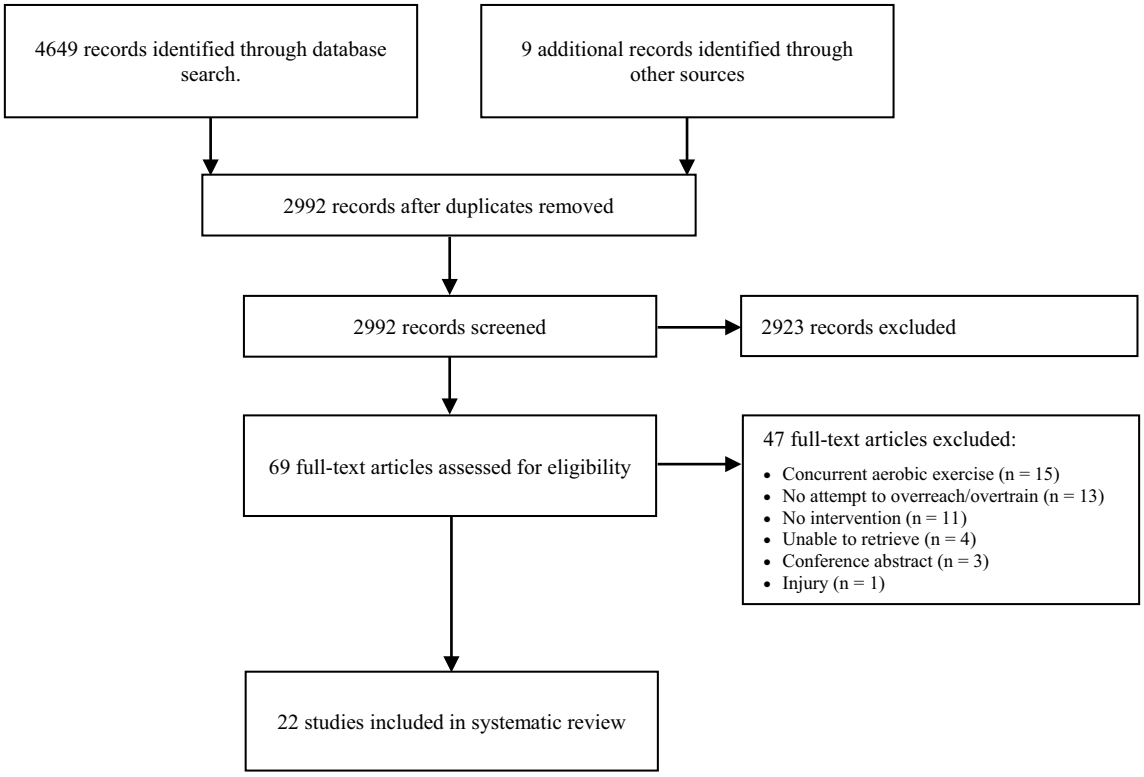

Fig. 1 PRISMA flow diagram of systematic search and included studies

results of the nine studies that did not adversely affect performance are displayed in the Table S4. Among the 12 studies that reported decreases in performance, nine reported on measures of biochemistry [11–13, 15–18, 20, 21], five studies investigated physiological changes [13, 16, 18–20], and three studies measured psychological and perceptual responses [9, 13, 18].

3.3 Performance

Twelve studies demonstrated decreases in at least one measure of performance following an overload training intervention (Table 2). Among these studies, eight implemented follow-up measures of performance. The majority of studies reported that performance was restored within the follow-up period. However, many studies failed to report the time course changes of performance within the follow-up period [9, 14, 15, 21]. The most common performance measure was maximal muscular strength. Six studies demonstrated significant decreases in strength [11–13, 16–18]. Contrasting results were found for secondary performance measures. For example, six studies measured vertical jump performance [9, 14–16, 18, 21], of these, three resulted in decreased performance [15, 18, 21]. Additionally, two studies reported decrements in sprint performance [9, 14]. Fry et al. [14] reported decreases in sprint performance; however, 1RM strength improved following the training intervention. Other secondary performance measures included in the analyses

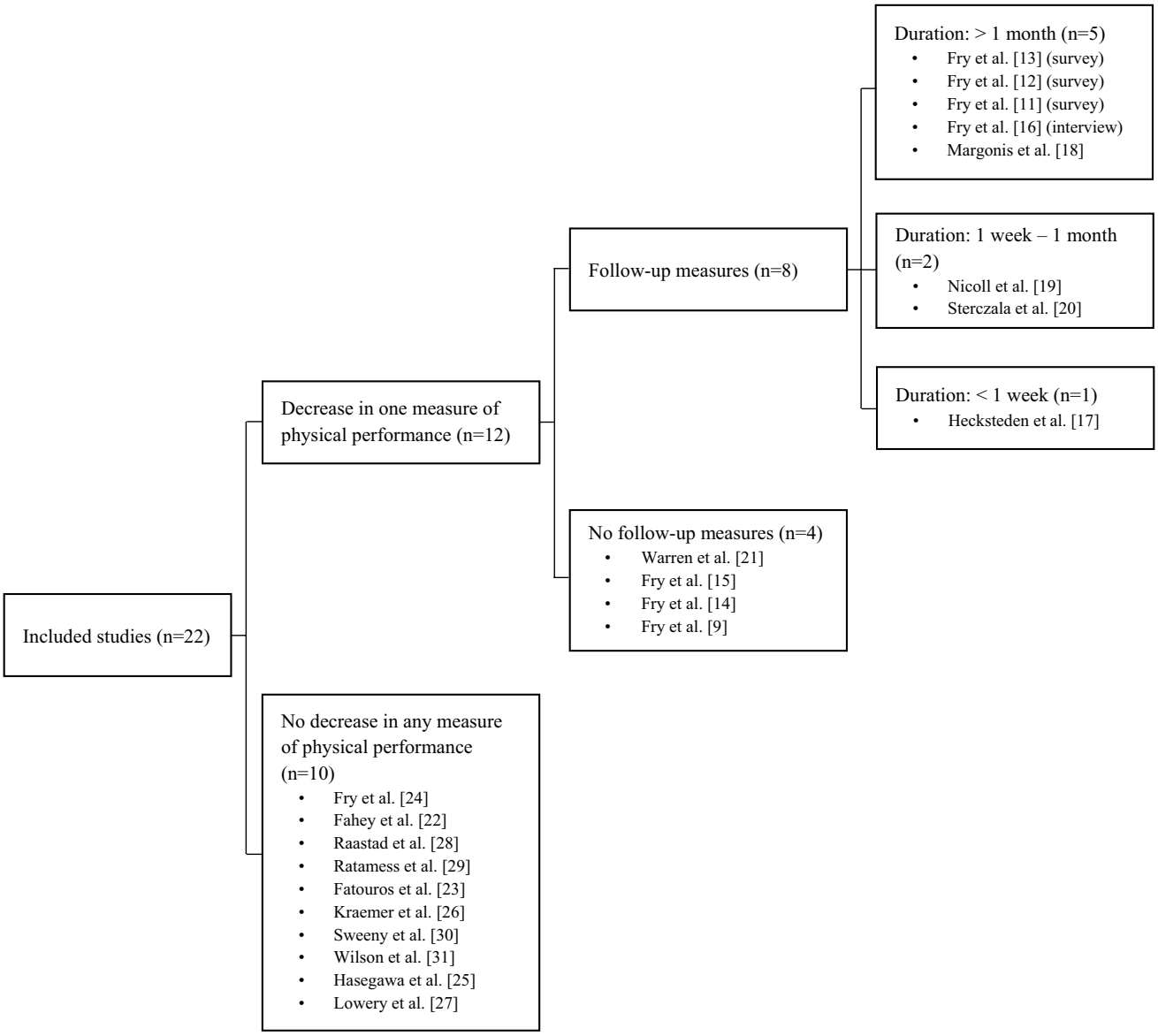

Fig. 2 Flow diagram of performance outcomes and follow-up interventions of included studies

Table 1 Descriptive results of studies that adversely affected at least one measure of physical performance

Study design	Study	Subjects	Overload training				Follow-up
			Duration	Frequency	Volume and intensity (sets × repetitions)	Mode	
Follow-up							
RCT	Fry et al. [13]	17 resistance trained, men	2 weeks	OT: 7 days/week C: 1 day/week	OT: 10 × 1 at 100% 1RM C: 3 × 5 at 50% 1RM	Tru-Squat machine (upper body resistance exercise held constant)	Follow up survey (8 weeks)
RCT	Fry et al. [12]	[13]	[13]	[13]	[13]	[13]	[13]
RCT	Fry et al. [11]	[13]	[13]	[13]	[13]	[13]	[13]
RCT	Fry et al. [16]	16 resistance trained, men	2 weeks	OT: 7 days/week C: 2 days/week	OT: 10 × 1 at 100% 1RM C: 3 × 5 @ 50% 1 RM	Tru-Squat machine	Follow up interviews (8 weeks)
UCT	Margonis et al. [18]	12 recreationally trained, men	3 weeks	6 days/week	6 × 1–6 at 85–100% 1RM	Bench press, squat, snatch, hang clean, deadlift, curl and row	3 weeks, 2 days/week: 2 × 10–12 at 70% 1RM Then 3 weeks rest
UCT	Nicoll et al. [19]	14 resistance trained, men HPOR ($n=6$) HIOT ($n=8$)	HPOR:7.5 days HIOT: [16]	HPOR: 2 sessions/day HIOT: [16]	HPOR: 10 × 5 at 70% 1RM (maximum concentric velocity) HIOT: [16]	HPOR: barbell back squat HIOT: [16]	HPOR: 1-week recovery HIOT: [16]
RCT	Sterczala et al. [20]	17 resistance trained, men AA ($n=8$) Placebo (OT) ($n=3$) Control ($n=6$)	1 week	AA/OT: 15 sessions C: maintained normal training schedule	AA/OT: 10 × 5 at 70% of the system 1RM mass. When barbell velocity dropped below 90% of the highest value for the set load was reduced. C: maintained normal training schedule	AA/OT: 'speed' barbell back squats C: maintained normal training schedule	1 week training cessation
UCT	Hecksteden et al. [17]	23 strength athletes, men and women	6 days	11 sessions	4 × 6 at 85% 1RM Implementing drop sets to failure and eccentric overload training	Bench press and squat (plus variations)	2 days rest
No follow-up							
UCT	Warren et al. [21]	28 elite junior weightlifters	7 days	2–3 sessions/day	Approximate volume: 90,000 kg/week Intensity: percentage of the participants best snatch and best clean and jerk	Snatch, clean, back squat, front squat, snatch pull and clean pull	

Table 1 (continued)

Study design	Study	Subjects	Overload training				Follow-up
			Duration	Frequency	Volume and intensity (sets × repetitions)	Mode	
RCT	Fry et al. [15]	28 male weightlifters AA ($n = 13$) Placebo ($n = 15$)	7 days	2–3 sessions/day	Approximate volume: 90,000 kg/day Intensity: 70–100% 1RM	Multi-joint barbell exercises: snatch, power snatch, snatch pull, clean, power clean, clean pull, jerk, push jerk, push press, clean and jerk, front squat, back squat	
RCT	Fry et al. [14]	9 resistance trained, men	3 weeks	OT: 5 days/week C: 2 days/week	OT: 8 × 1 at 95% 1RM C: 3 × 5 at 70% body weight	Tru-Squat machine	
NRCT	Fry et al. [9]	6 resistance trained, men	3 weeks	3 days/week	Barbell back squat: 2 × 1 at 95% 1 RM 3 × 1 at 90% 1RM Leg curls: 3 × 10RM	Barbell back squat and leg curls (upper body resistance exercise held constant)	

RCT randomised controlled trial, *UCT* uncontrolled trial, *RM* repetition maximum, *OT* overload training, *C* control, *NR* not reported, *HMB-FA* beta-hydroxy-beta-methylbutyrate-free acid, *AA* amino acid, *HPOR* high power overreaching, *HIOT* high-intensity overtraining, *kg* kilogram

were maximal repetitions until exhaustion [13, 15, 21], Wingate power [18], sit and reach [9], and muscular torque [13, 14].

3.4 Biochemistry

Biochemical responses were reported by nine studies that demonstrated decreases in performance. A variety of blood and urinary biomarkers including hormones and muscle damage markers as well as oxidative stress and inflammatory markers were reported. Four studies reported on the between-group biochemical responses of overload participants in comparison to a control (Table 3). Additionally, nine studies provided pre- to post-intervention results for overload training participants (Table S5). Commonly reported measures included cortisol [11, 15, 20], testosterone/free

Table 2 Performance results of studies that adversely affected at least one measure of physical performance

Study	Study design	Performance post overload training			Follow-up
		Improved	No change	Declined	
Fry et al. [13]	RCT		Squat repetitions @ 70% 1RM	1RM squat Isokinetic leg extension torque	Follow-up survey indicated that participants were able to return to normal training load with 2–8 weeks of recovery.
Fry et al. [12]	RCT		[13]	[13]	[13]
Fry et al. [11]	RCT		[13]	[13]	[13]
Fry et al. [16]	RCT		Vertical jump (CMJ, squat jump, depth jump) Mean power, peak velocity and peak force @ 40, 70% 1RM	1RM squat Mean power @ 100% 1RM	Follow-up interviews indicated that participants were able to return to normal training load with 2–8 weeks of recovery.
Margonis et al. [18]	UCT			1RM strength Vertical jump Wingate power	1RM strength returned to baseline values following 3 weeks of recovery.
Nicoll et al. [19]	UCT		HIOT: [16]	HPOR: squat mean power HIOT: [16]	HPOR: muscular power returned to baseline following 1 week of recovery. HIOT: [16]
Sterczala et al. [20]	RCT		1RM back squat Squat mean power @70% 1RM	Squat mean barbell velocity @ 70% 1RM	Mean barbell velocity values returned to baseline following 1 week of training cessation
Hecksteden et al. [17]	UCT			Squat maximal isometric contraction Bench press maximal isometric contraction	Performance was not restored to baseline values following 2 days rest
Warren et al. [21]	UCT		Snatch lift test to exhaustion	Vertical jump	
Fry et al. [15]			Snatch (repetitions, missed lifts & successful lifts)	Vertical jump	
Fry et al. [14]	RCT	1RM squat Agility run (left)	Isokinetic leg extension torque Agility run (right) Vertical jump (CMJ, non-CMJ, low depth, high depth)	Sprint	
Fry et al. [9]	UCT		1RM squat Sit and reach Lower body reaction time Vertical jump Back and hip extension isokinetic peak force	Sprint Squat isokinetic peak force	

RCT randomised controlled trial, *NRCT* non-randomised controlled trial, *UCT* uncontrolled trial, *HPOR* high power overreaching, *HIOT* high-intensity overtraining, *RM* repetition maximum, *CMJ* countermovement jump

testosterone [11, 15, 20], testosterone:cortisol (T:C) ratio/free T:C ratio [11, 15, 20], basal growth hormone [11, 15, 17] and exercise induced growth hormone [11, 15], catecholamines [12, 16, 20], creatine kinase (CK) [13, 17] and pre- and post-exercise blood lactate/lactic acid [12, 13, 15, 21]. In comparison to a control, overload training participants demonstrated exacerbated responses of cortisol [20], creatine kinase [13] and noradrenaline [12]. Blunted responses were reported in T:C and free T:C ratio [11] and exercise-induced lactate concentrations [13].

3.5 Physiology

Amongst the studies that demonstrated declines in performance, five measured physiological adaptations. For example, Fry et al. [13] reported no significant changes in resting heart rate upon waking. Decreases in range of motion were demonstrated following 3 weeks of overload resistance training [18]. Three studies investigated the physiological mechanisms that may underpin the decline in performance characteristic of overtraining [16, 19, 20]. No significant changes were demonstrated in contractile protein isoform expression [16, 19]. Fry et al. [16] and Sterczala et al. [20] reported downregulation of the β_2-adrenergic receptor (β_2-AR) system with simultaneous increases in epinephrine:β_2-AR density following overload training. Additionally, Nicoll et al. [19] reported altered mitogen-activated protein kinase (MAPK) activity in response to overtraining in resistance exercise.

Table 3 Between-group biochemical responses of overload participants in comparison to control participant following the completion of overload training

Parameter	Measure	Blunted response	No significant difference	Elevated response
Hormones				
Cortisol	R			[20]
Testosterone	R		[20]	
T:C ratio	R	[11]	[20]	
	EI		[11, 20]	
Free T:C ratio	R	[11]		
	EI		[11]	
Adrenaline	R		[12, 20]	
	EI		[12]	
Noradrenaline	R		[12]	
	EI			[12]
Adrenaline: β_2-AR density	R			[20]
Blood biomarkers and muscle damage				
Lactate	R		[13]	
	EI	[13]		
Creatine kinase	R			[13]

T:C testosterone:cortisol, *β_2-AR* β_2-adrenergic receptor, *R* resting, *EI* exercise induced

3.6 Psychological and Perceptual Responses

Psychological and perceptual responses were reported by three studies that demonstrated declines in performance. Margonis et al. [18] demonstrated increases in delayed onset muscle soreness following the increase in training load. Conflicting results were reported for perceived lower back pain and knee pain. No increases in pain were reported where participants trained 3 days per week [9]. However, increases in pain were demonstrated in a study where participants trained 7 days per week [13]. Fry et al. [13] also reported decreases in perceived recovery and strength status following the overload training intervention. However, no significant changes in sleep pattern (sleep quality and duration) were reported throughout the trial [13].

3.7 Risk of Bias

The risk of bias assessment results, including selection bias, performance bias and confounding bias, is available in Fig. S6. In the case of UCTs, selection bias was assessed as not applicable as study protocol did not involve allocation of participants. No studies demonstrated adequate allocation concealment. Random sequence generation was performed by one study [31]. Blinding of participants was adequately performed by seven studies [15, 20, 22, 26–28, 31]. However, as these studies involved supplementation groups, only the results of the placebo participants are relevant for discussion in this review. A total of eight studies are reflective of possible FOR, NFOR or OTS [11–13, 16–20]. These studies demonstrated an adequate decrease in at least one measure of performance and included follow-up measures after the overload period. Of these studies, only one adequately controlled for nutritional confounders [18]. An additional bias may exist as the majority of the studies included in this review came from the same group of authors.

4 Discussion

Despite the fact that many athletes perform resistance-based exercise as a significant part of their training, much of the scientific literature on overtraining has focused on endurance activities [7]. This review analysed changes and attempted to identify relationships between numerous measures of performance, biochemistry, physiology and psychology that occurred in response to FOR, NFOR or OTS in resistance exercise. This systematic review also aimed to develop an understanding of the mechanisms and manifestation of overtraining in resistance exercise. Such findings may provide

an early indication of training maladaptation. However, to accurately elucidate the effects of overtraining in resistance exercise, it is important to accurately diagnosis a state of FOR, NFOR or OTS in intervention participants.

4.1 Summary of Evidence

4.1.1 Establishing Overreaching/Overtraining

Previous studies have attempted to investigate the time course changes of training maladaptation in resistance exercise. However, many investigations have failed to appropriately induce and diagnose FOR, NFOR or OTS. Methodological limitations and contrasting definitions within the literature contribute to the discrepant findings. At present, changes in performance and recovery time are required to diagnose and distinguish between these conditions [7]. Amongst the 22 studies included in this review, only eight studies reported declines in performance and implemented follow-up measures. However, few studies used a longitudinal approach to monitor performance during a follow-up/recovery period. Due to the short duration and lack of performance testing throughout the follow-up period, the time course changes of performance in much of the resistance exercise overtraining literature remain unknown. Additionally, many studies that adversely affected performance did not measure performance during a follow-up period. Whilst these studies may provide some insight into training maladaptation, it is not possible to determine the prognosis of training maladaptation if performance is not adequately monitored throughout a follow-up period.

Many of the previous resistance overtraining interventions failed to adversely affect performance following an overload training period. Whilst these studies are not reflective of a state of FOR, NFOR or OTS, many researchers reported that overreaching or overtraining occurred. For example, Raastad et al. [28] did not report any decreases in performance measures (i.e. muscular strength), yet suggested that this should be recognised as overreaching as the strength gain per workout was reduced. However, a decline in strength adaptation per training bout may simply reflect the law of diminishing returns rather than a maladaptation to training. Future studies should attempt to standardise methodology (e.g. follow-up period and performance testing) and diagnostic criteria (e.g. demonstrated decrease in performance lasting 'days-weeks' for FOR, 'weeks-months' for NFOR and 'months…' for OTS) to avoid inappropriate determination of FOR, NFOR and OTS in resistance training.

4.1.2 Performance

Performance outcomes of resistance training have been measured using a number of different tests including maximal muscular strength, muscular torque, force production and maximal repetitions until exhaustion [9, 16, 21]. To establish training maladaptation, the performance test utilised should be a reflection of an athlete's competition performance [7]. However, there is a lack of consistency in performance measures implemented within studies. This may be due to the diverse nature of resistance training and the variety of athletic performance outcomes (e.g. maximal strength, muscular power or hypertrophy). If the performance outcome of a resistance-based sport/discipline is not maximal strength, it is counterintuitive to measure maximal strength as a primary performance indicator. Therefore, future studies utilising resistance training must specify a primary performance test reflective of the demands and environmental context of an athlete's competition performance.

The most common methods for assessing performance in included studies are maximal muscular strength tests. Maximal muscular strength tests are a resistance training-specific tool that can be utilised for the detection of FOR, NFOR or OTS in resistance exercise. However, maximal strength tests are exhaustive in nature and may contribute to fatigue levels which increase the risk of training maladaptation. Few studies demonstrated decreases in maximal strength. For example, Margonis et al. [18] reported significant decreases in 1RM strength which were not restored following a 6-week taper period. These results suggest that athletes developed the OTS. Hecksteden et al. [17] reported a significant decline in maximal isometric force that did not return to normal values following a recovery period of 2 days. However, due to the short follow-up period the length of recovery required to distinguish between FOR, NFOR or OTS, correct diagnosis is not possible. Additional studies that reported decreases in maximal strength utilised participant interviews and surveys to monitor the length of recovery required to re-establish performance [11–13, 16]. Collectively, these studies have reported attenuated performance which may provide important insight into the mechanisms and manifestation of training maladaptation. However, as these studies did not appropriately measure performance during a sufficient follow-up period it is not possible to determine if these results reflect FOR, NFOR or OTS. Future research should implement follow-up measures to assess the time course changes of performance to appropriately diagnose FOR, NFOR or OTS.

The lack of longitudinal interventions which have demonstrated training maladaptation presents difficulties in the attempt to monitor and diagnose overtraining. It appears as though maximal muscular strength is resilient to large acute increases in training load. Therefore, secondary performance measures may be useful for the early detection of FOR,

NFOR or OTS in resistance-trained athletes. Previous studies have employed a variety of performance measures (e.g. sprint time, vertical jump height) not specific to the resistance training protocol that may provide important insight into the monitoring of possible overtraining [14–16, 21]. Contrasting results were reported for secondary performance measures. Collectively, it appears as though changes in maximal strength may not reflect performance in other tasks. Future studies should aim to appropriately establish FOR, NFOR or OTS and determine whether such tests could be used to monitor the manifestation of these conditions. Performance testing is an efficient and inexpensive method to monitor and prevent overtraining. However, given the short duration and wide variety of performance measures used by many studies, there remains insufficient evidence to reach a consensus on performance tests that monitor overtraining in resistance exercise.

The majority of the literature on overtraining in resistance exercise has been reported following interventions where physical performance was not adversely affected. Although it is difficult to compare without consistent measures between studies, differences in the prescription of overload training protocols may provide insight into the type of resistance training that may increase susceptibility to overtraining. The training frequency and relative training intensity of interventions that did induce performance decrements appear to be higher than those that did not adversely affect performance. All studies that performed multiple sets of one repetition at high intensity (e.g. 10 sets × 1 repetition at 100% 1RM, every day for 2 weeks) were able to induce performance decreases [11–13, 16]. Low volume, high-intensity and -frequency protocols such as those used in these interventions may be insightful for research purposes. However, coaches and athletes are unlikely to implement such regimes as part of a training program due to the risk of injury associated with training at maximal loads [33]. Previous studies that demonstrated no adverse effects on performance implemented training protocols that are likely more reflective of how resistance-based athletes would normally train. However, as many overtraining studies have been unable to induce FOR, NFOR or OTS and the concomitant performance decrements, the trade-off between ecological validity and adequate manipulation of exercise volume and/or intensity to elicit a maladaptive response must be considered. These studies may still provide important insight into resistance exercise FOR, NFOR and OTS.

Training monotony may be expressed as the amount of variation in training load within one training cycle [34]. A previous study has suggested that monotonous training increases susceptibility to overreaching and overtraining in endurance activities [35]. Although there is little empirical evidence examining the relationship between training monotony and overtraining, it appears as through monotony of training load may increase the likelihood of overtraining in resistance exercise. Many previous studies that reported performance decrements utilised interventions where only one type of resistance exercise was performed (i.e. squat) [11–14, 16, 19, 20]. It appears as though low variation in exercise selection or targeted muscle groups may increase the likelihood of training maladaptation due to a high increase in concentrated training load with limited time to recover. Therefore, coaches and athletes should use caution when programming an overload period targeting a specific lift or muscle group. Collectively, high training frequency, high training intensity and training monotony appear to increase the likelihood of training maladaptation. However, confounding factors such as training status must be considered in future studies.

4.1.3 Biochemistry

Various biochemical markers have also been investigated to gain a greater understanding of the potential mechanisms of overtraining in resistance exercise. Nine studies measured biochemical responses following an overload training period that resulted in a decrease in performance [11–13, 15–18, 20, 21]. Marked responses were observed in overload participants in comparison to control participants in hormone levels (cortisol, T:C ratio, free T:C ratio and catecholamines) [11, 12, 20], blood biomarkers and muscle damage markers (lactate and creatine kinase, respectively) [13]. However, the biochemical responses of overtraining remain unknown due to the variability of results and lack of studies that have measured the time course changes in performance required to accurately diagnose maladaptive training conditions. A recent systematic review of the hormonal aspects of overtraining by Cadegiani et al. [36] also recognised that the improper diagnosis of FOR, NFOR and OTS is a significant problem in overtraining literature. Furthermore, biochemical markers have been unable to distinguish between a normal training response, acute fatigue, FOR, NFOR and OTS. Further research is needed to understand the biochemical alterations during training maladaptation and to determine the validity and reliability of biomarkers for monitoring overtraining. It has previously been suggested that overtraining can result in differential biochemical responses depending on whether high relative intensity or high-volume resistance training is performed [8, 37]. However, due to the lack of studies that have appropriately established overtraining through high-volume and/or high-intensity training protocols, comparisons cannot be made; therefore, this theory cannot be corroborated.

4.1.4 Physiology

Physiological measures have previously been investigated as tools to monitor and/or diagnose overtraining. However, relatively few studies have investigated the response of physiological measures to overtraining in resistance exercise. Fry et al. [13] measured resting heart rate upon waking and reported no changes throughout 2 weeks of overload resistance training. However, the multifactorial causes of variation in basal heart rate may limit its ability to monitor for overtraining in resistance exercise. Additionally, Margonis et al. [18] reported on knee range of motion as a measure of the swelling/oedema response to overtraining in resistance exercise. The authors reported a decrease in range of motion indicative of swelling and muscle micro trauma following the overload training intervention. However, inflammatory measures may be unable to distinguish between acute fatigue, FOR, NFOR and OTS. The muscle physiology mechanisms that underpin the decline in performance characteristic of overtraining in resistance exercise have been investigated by relatively few studies [16, 19, 20]. Skeletal muscle fibre composition and muscle fibre contractile properties have previously been examined. Declines in β_2-AR density and altered MAPK activity may be responsible for the muscle maladaptation associated with overtraining. However, future research is needed to determine the validity and reliability of physiological measures as tools to monitor overtraining in resistance exercise. Furthermore, physiological monitoring markers must reflect the nature and mechanisms of resistance training.

4.1.5 Psychology and Perceptual Responses

It is well established that the presence of negative psychological and perceptual responses coincides with overtraining in endurance populations [7]. However, relatively few studies have used psychological and perceptual monitoring tools in resistance exercise overtraining interventions. Increases in delayed onset muscle soreness and pain as well as decreases in perceived recovery status and perception of strength have been reported among interventions that demonstrated a decrease in performance [13, 18]. These are expected findings given that the process of overtraining involves an imbalance of training stress and recovery. In contrast to the endurance literature, Fry et al. [13] reported slight decreases in anxiety following an overload training period which resulted in decreases in 1RM strength. The authors suggested that overtraining through high-intensity resistance exercise may produce different psychological responses in comparison to high-volume endurance training. At present, perceptual measures of soreness and recovery status may provide information concerning increases in training load. However, these tools are yet to be established as reliable indicators of overtraining in resistance exercise. Due to the lack of studies demonstrating overtraining and investigating psychological and perceptual responses to overtraining in resistance exercise, no clear consensus can be reached. Future investigators should combine tools to measure psychological disturbances with measures of performance to gain a better understanding of this relationship.

4.2 Limitations

This is the first systematic review to critically appraise the methodology and results of studies that have attempted to induce FOR, NFOR or OTS through resistance exercise. This analysis was limited by the inability to perform a meta-analysis due to the heterogeneity of study methodologies and outcomes. Additionally, a possible risk of bias may exist in the selection of search terms and exclusion criteria as defined by the authors. However, this review followed protocols as recommended by the AMSTAR 2 and PRISMA guidelines. Furthermore, due to the criteria required to diagnose overtraining, an adapted risk of bias assessment tool was required to accurately reflect the quality of studies. Limitations are present in this assessment as a clear judgement about the risk of bias arising from each domain was not reflected with a validated assessment tool (i.e. Cochrane risk of bias appraisal tool). To address this limitation, an adapted risk of bias assessment tool was used. Additionally, a variety of limitations exist in the included studies. Many studies did not demonstrate decreases in performance or measure the time course changes in performance during a follow-up period. Due to these methodological limitations, an accurate diagnosis of FOR, NFOR or OTS was impossible. The lack of standardised diagnostic criteria and the inconsistency and variability in measurements of performance, biochemistry, physiology and psychology amongst studies make it difficult to establish the relationship between overtraining and resistance exercise.

4.3 Future Research

Given the negative impact of overtraining on athletic performance, it is essential that research in this area is examined critically. Due to the lack of studies demonstrating decreases in performance and implementing follow-up measures, further research is required to develop an understanding of overtraining in resistance exercise. The prevalence and symptoms of overtraining in resistance exercise should be further investigated given the lack of longitudinal studies in resistance-trained populations. Future studies must implement resistance training interventions that are reflective of the sport/discipline to maintain ecological validity; however, the authors should report the primary performance measure. To gain a better understanding of the effects of overtraining,

future studies should attempt to increase consistency amongst measured parameters (biochemical, physiological and psychological markers). There is also a need to investigate the time course changes in performance, biochemistry, physiology and psychology throughout the recovery/follow-up stage. Future investigators should also consider examining the aetiology of high-volume versus high relative intensity resistance exercise overtraining. Additionally, future authors should control for confounding factors (i.e. nutritional and sleep characteristics) as these factors may play a role in the development of the OTS [38]. As was recently done in an overtraining study by Cadegiani et al. [38], future authors should consider utilising the flowchart proposed by Meeusen et al. [7] to diagnose OTS and account for confounding factors.

It is necessary to consider the clinical and economic implications of studying overtraining in athletic populations. Future researchers must consider the impact of subjecting participants to increased training loads over long periods in an attempt to induce overtraining. This may result in long-term decrements in performance and increased susceptibility to physical or psychological illness and injury. Additionally, overtraining may be experienced differently amongst participants. Future studies must consider interindividual variability in response to increases in training load. While the difficulties and ethical considerations of inducing overtraining in athletic populations are appreciated, well-designed studies with demonstrated decreases in performance and follow-up measures are required to develop a greater understanding of overtraining in resistance exercise.

5 Conclusion

The primary aim of this review was to critically analyse the existing literature to elucidate the mechanisms that underlie overtraining in resistance exercise. Despite the potentially serious implications of overtraining in athletic populations, to date, there is insufficient literature investigating the effects of FOR, NFOR and OTS in resistance exercise. The majority of previous resistance-based overtraining interventions fail to demonstrate decrements in training-specific performance. Furthermore, many studies do not accurately measure the length of recovery and time course changes in physical performance. It is apparent that the misdiagnosis of overtraining is a significant problem in previous research. However, it is necessary to mention that whilst many studies did not appropriately establish FOR, NFOR or OTS they still provide valuable information regarding the physiological alterations and potential mechanisms of resistance training fatigue.

At present, there is insufficient evidence to draw accurate conclusions on the effects of overtraining in resistance exercise. No diagnostic tool other than a decrease in sport-specific performance can be taken as an indicator of overtraining in resistance exercise. Given the current evidence the exact role of biochemical, physiological and psychological markers in overtraining in resistance exercise remains equivocal. Athletes and coaches should regularly monitor a combination of performance, biochemical, physiological and psychological variables to identify athletes that may be approaching training maladaptation. The causes and pathogenic nature of overtraining remain unclear; however, frequent, high-intensity, monotonous resistance training appears to increase an athlete's susceptibility to overtraining. However, caution must be exercised when interpreting these findings due to the lack of interventions which have induced FOR, NFOR or OTS through resistance exercise.

Data availability statement All data generated or analysed during this study are included in this published article (and its supplementary information files).

Compliance with ethical standards

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